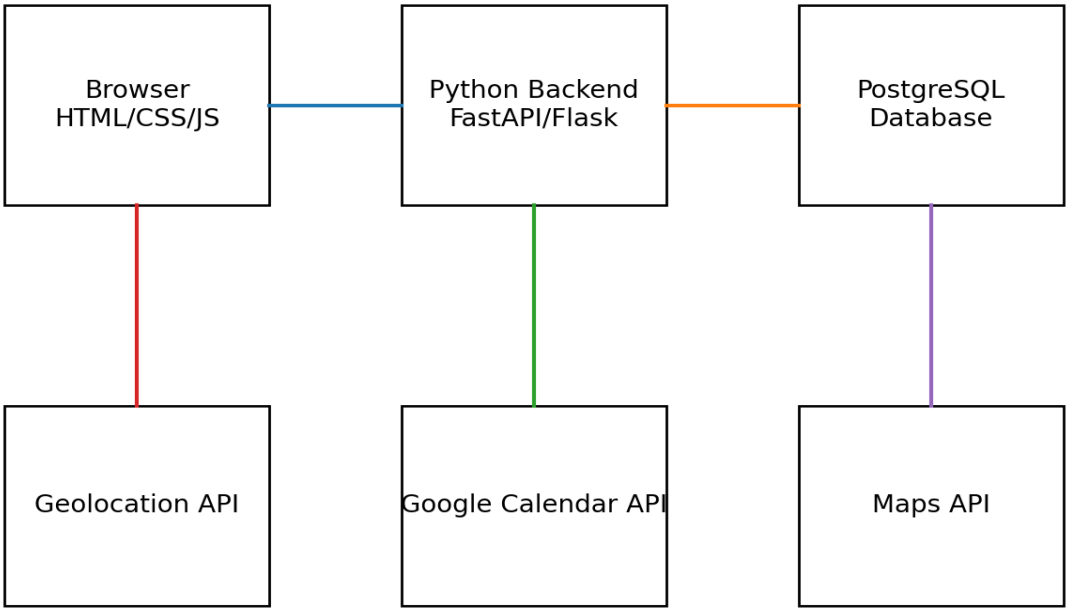
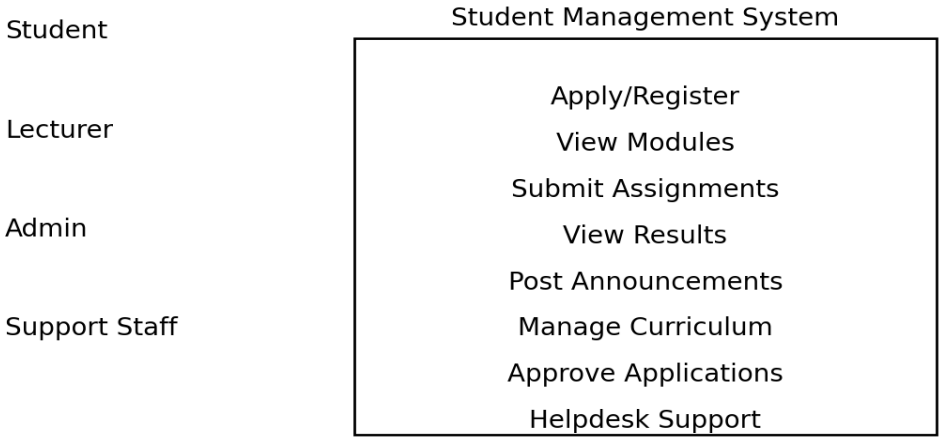
Student Management System - Architecture & UML Documentation

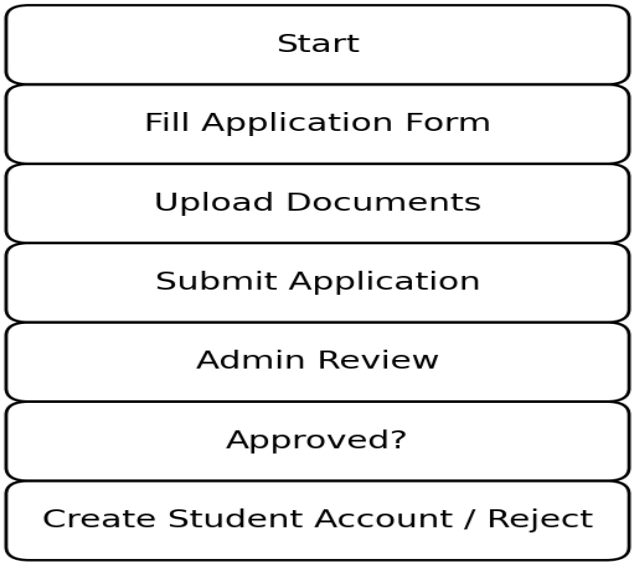
System Architecture Diagram



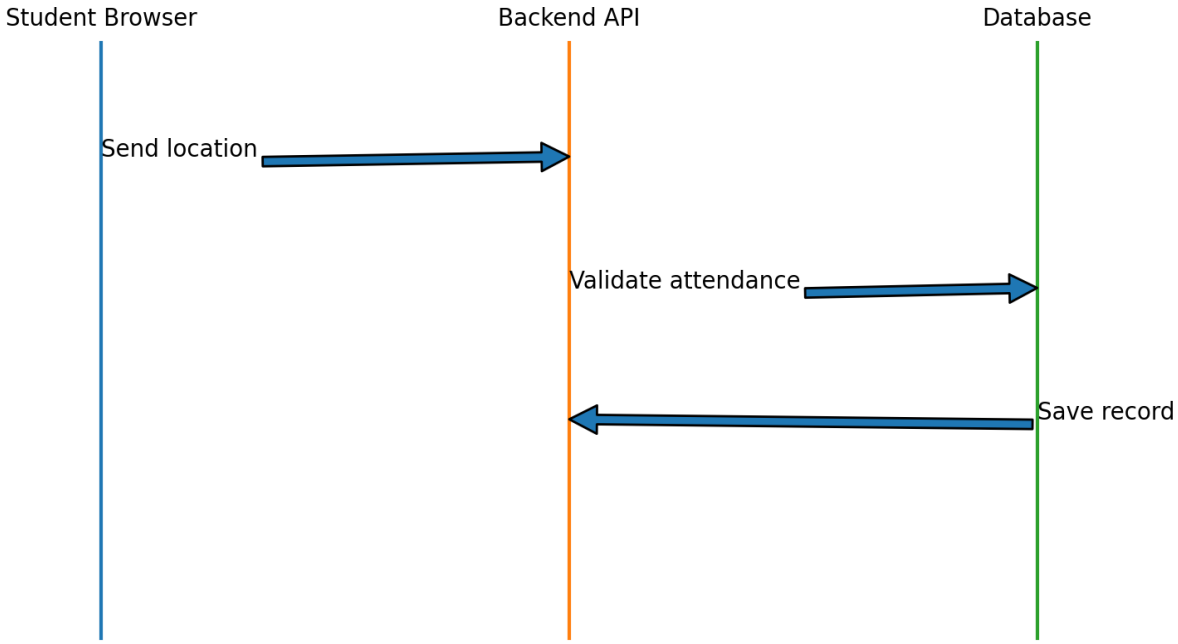
Use Case Diagram



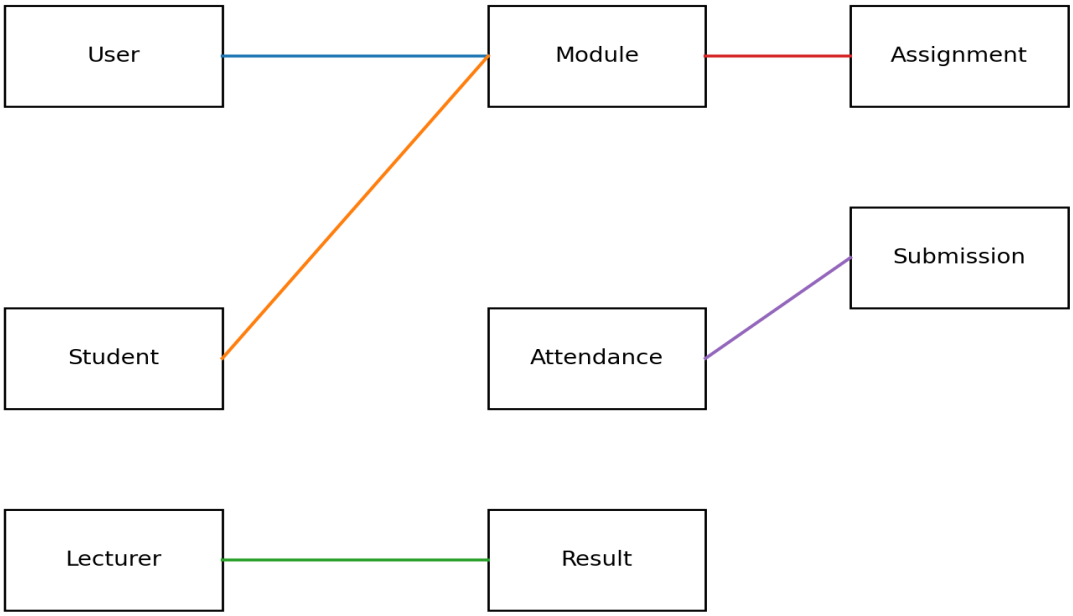
Activity Diagram



Sequence Diagram



Entity Relationship Diagram



Database Schema

```
DATABASE SCHEMA (Core Tables)

users(id, email, password_hash, role, created_at)
students(id, user_id, sector, funded, funding_amount, application_status)
lecturers(id, user_id, department)
modules(id, code, title, sector, credits)
enrollments(id, student_id, module_id)
assignments(id, module_id, title, due_date)
submissions(id, assignment_id, student_id, file_path, grade)
attendance(id, student_id, module_id, timestamp, lat, lng)
results(id, student_id, module_id, score, grade)
announcements(id, author_id, title, body)
tickets(id, student_id, department, subject, status)
suppliers(id, name, service_category, contact)
venues(id, name, building, lat, lng)
timetable(id, module_id, venue_id, start_time, end_time)
```

Full System File Structure

```
SYSTEM FILE STRUCTURE

sms-system/
■
■■■ backend/
■   ■■■ app.py
■   ■■■ config.py
■   ■■■ models/
■   ■   ■■■ user.py
■   ■   ■■■ student.py
■   ■   ■■■ lecturer.py
■   ■   ■■■ module.py
■   ■   ■■■ assignment.py
■   ■   ■■■ attendance.py
■   ■   ■■■ results.py
■   ■
■   ■■■ routes/
■   ■   ■■■ auth_routes.py
■   ■   ■■■ student_routes.py
```

```
■   ■   ■■■ lecturer_routes.py
■   ■   ■■■ assignment_routes.py
■   ■   ■■■ attendance_routes.py
■   ■   ■■■ helpdesk_routes.py
■   ■
■   ■■■ services/
■   ■   ■■■ calendar_service.py
■   ■   ■■■ attendance_service.py
■   ■   ■■■ email_service.py
■   ■
■   ■■■ database/
■       ■■■ migrations/
■
■■■ frontend/
■   ■■■ templates/
■   ■   ■■■ dashboard.html
■   ■   ■■■ modules.html
■   ■   ■■■ timetable.html
■   ■   ■■■ assignments.html
■   ■   ■■■ helpdesk.html
■   ■
■   ■■■ static/
■       ■■■ css/
■       ■■■ js/
■       ■■■ images/
■
■■■ docs/
■   ■■■ uml/
■   ■■■ architecture/
■   ■■■ api/
■
■■■ tests/
■
■■■ README.md
```

GitHub Project Board Structure

```
GITHUB PROJECT BOARD STRUCTURE

Columns

Backlog
Ready
In Progress
Code Review
Testing
Done

Example Issues

#1 Setup backend framework
#2 Setup database schema
#3 Implement authentication
#4 Student application module
#5 Attendance geolocation module
#6 Assignment submission module
#7 Helpdesk ticket system
#8 Supplier management
#9 Google Calendar integration
#10 Campus map navigation
```